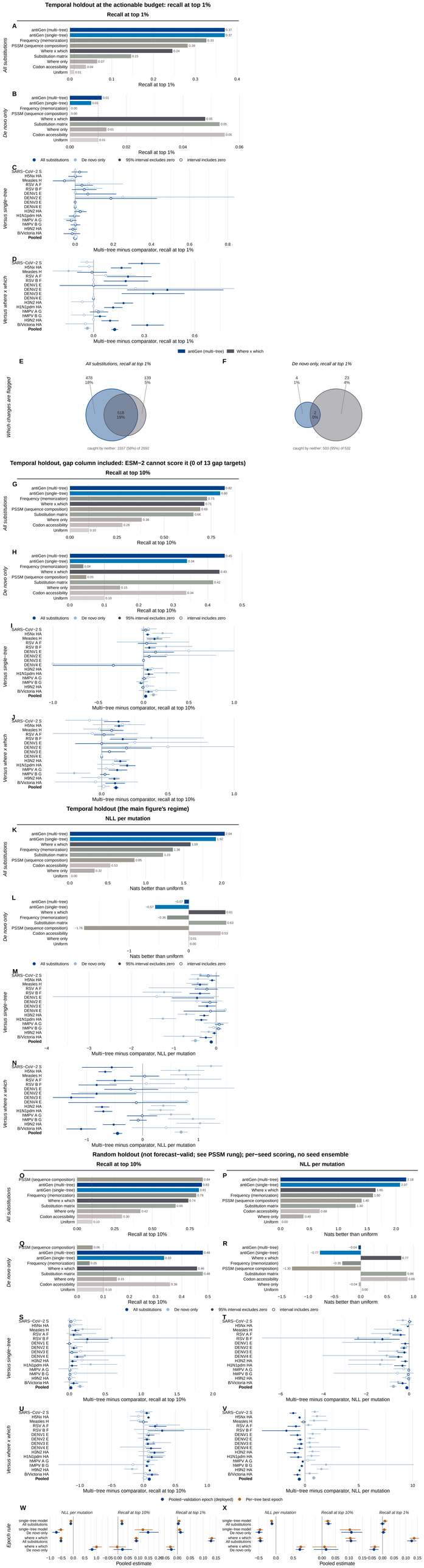
Temporal holdout at the actionable budget: recall at top 1%
Recall at top 1%
All substitutions
De novo only
Versus single−tree
Versus where x which
Which changes are flagged
antiGen (multi−tree)
antiGen (single−tree)
Frequency (memorization)
PSSM (sequence composition)
Where x which
Substitution matrix
Where only
Codon accessibility
Uniform
Recall at top 1%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV A G
hMPV B G
H9N2 HA
BVictoria HA
Pooled
Multi−tree minus comparator, recall at top 1%
Where x which
All substitutions, recall at top 1%
De novo only, recall at top 1%
caught by neither: 1557 (58%) of 2692
caught by neither: 503 (95%) of 532
Temporal holdout, gap column included: ESM−2 cannot score it (0 of 13 gap targets)
Recall at top 10%
Multi−tree minus comparator, recall at top 10%
Temporal holdout (the main figure's regime)
NLL per mutation
Nats better than uniform
Multi−tree minus comparator, NLL per mutation
Random holdout (not forecast−valid; see PSSM rung); per−seed scoring, no seed ensemble
Pooled−validation epoch (deployed)
Per−tree best epoch
NLL per mutation
Recall at top 10%
Recall at top 1%
Epoch rule
single−tree model
All substitutions
De novo only
Where x which
Pooled estimate